

Supplementary Materials

Ancient genomes reveal the genetic history of domestic goats on the Qinghai-Xizang Plateau approximately 3 600 years ago

Tian-You Bai^{1,#}, Ya-Hui Qiu^{2,#}, Hai-Feng He^{1,#}, Hong Wang³, Le Tao⁴, Li-Min Qiu¹, Kong-Yang Zhu¹, Yu Xu¹, Rui Wang⁵, Jia-Jing Zheng¹, Xiao-Min Yang⁶, Qu Shen⁶, Hao Ma¹, Jing Zhao^{7,*}, Pei-Xian Shu^{8,9,*}, Chuan-Chao Wang^{1,4,5,6,*}

¹School of Life Sciences, Xiamen University, Xiamen, Fujian 361102, China

²Shaanxi Academy of Archaeology, Xi'an, Shaanxi 710109, China

³Interdisciplinary Research Center of Earth Science Frontier, Beijing Normal University, Beijing 100875, China

⁴Institute of Forensic Science, Fudan University, Shanghai 200032, China

⁵Ministry of Education Key Laboratory of Contemporary Anthropology, Center for Evolutionary Biology, Department of Anthropology and Human Genetics, School of Life Sciences, Fudan University, Shanghai 200433, China

⁶Fujian Provincial Key Laboratory of Philosophy and Social Sciences in Bioanthropology, Institute of Anthropology, Xiamen University, Xiamen, Fujian 361005, China

⁷College of Environment and Life Sciences, Weinan Normal University, Weinan, Shaanxi 714099, China

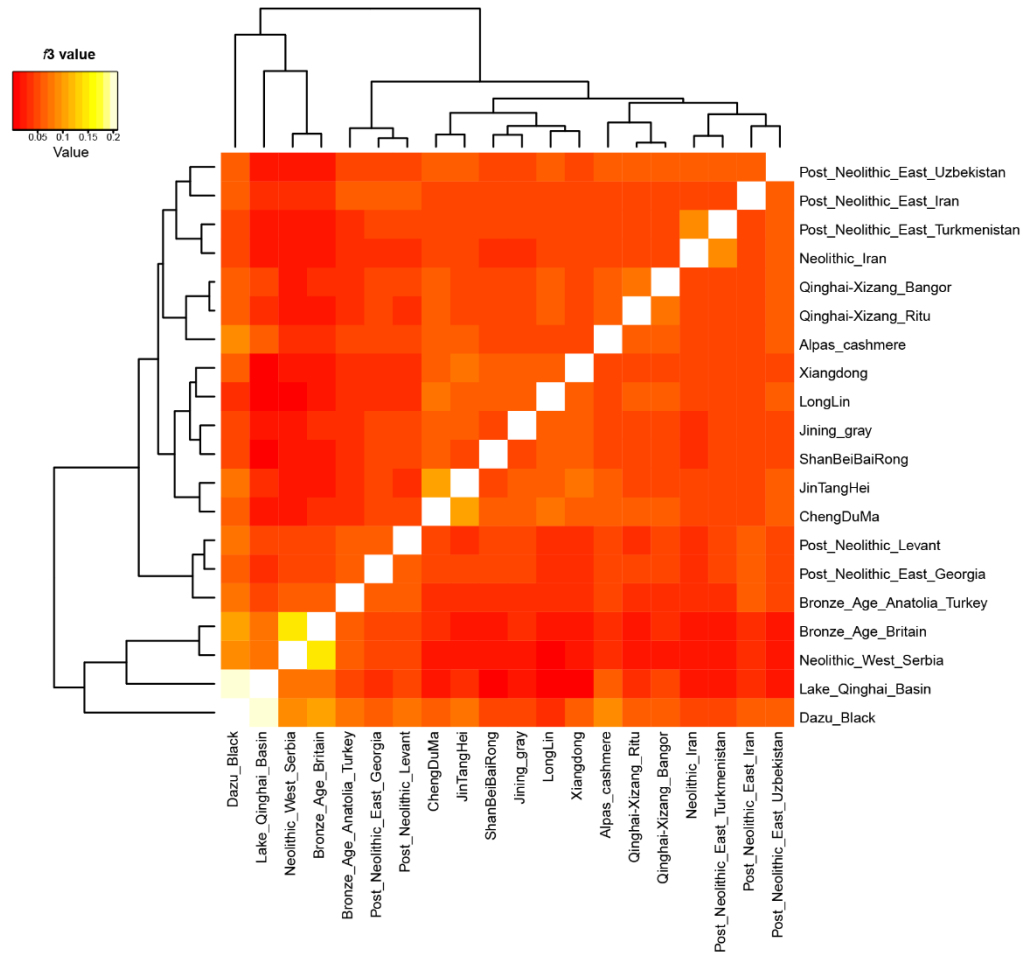
⁸State Key Laboratory of Loess Science, Institute of Earth Environment, Chinese Academy of Sciences, Xi'an, Shaanxi 710061, China

⁹Shaanxi Key Laboratory of Accelerator Mass Spectrometry Technology and Application, Xi'an AMS Center, Xi'an, Shaanxi 710061, China

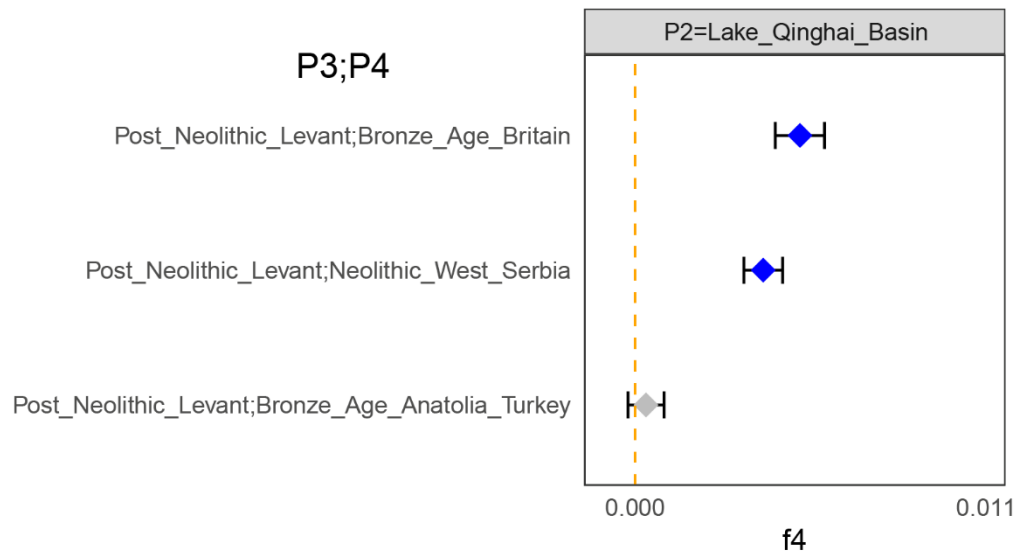
[#]Authors contributed equally to this work

^{*}Corresponding authors, E-mail: 465582984@qq.com; shupx@ieccas.cn; wang@xmu.edu.cn

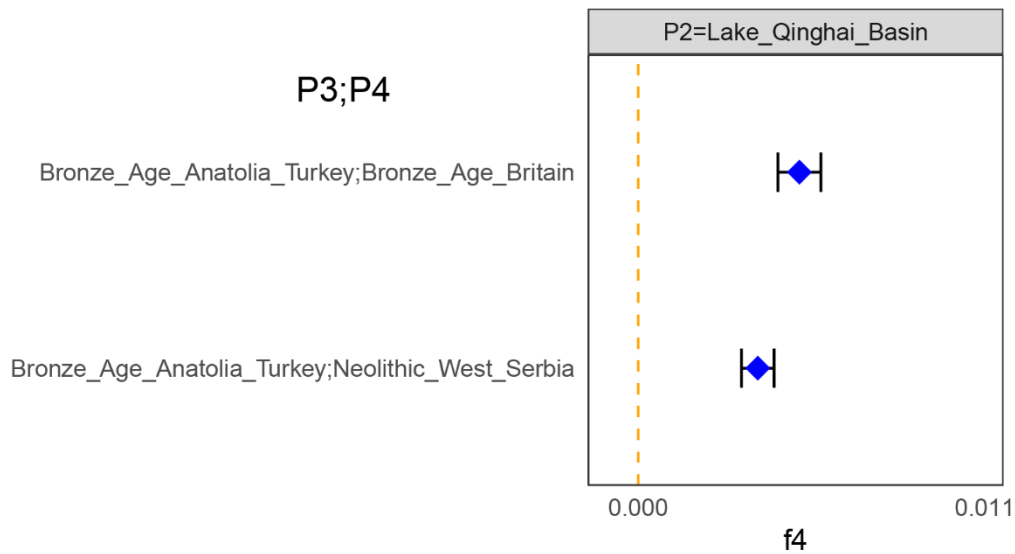
for $K=2$ to $K=4$. yellow component maximized were dominantly distributed in Lake Qinghai Basin and Dazu black goat.



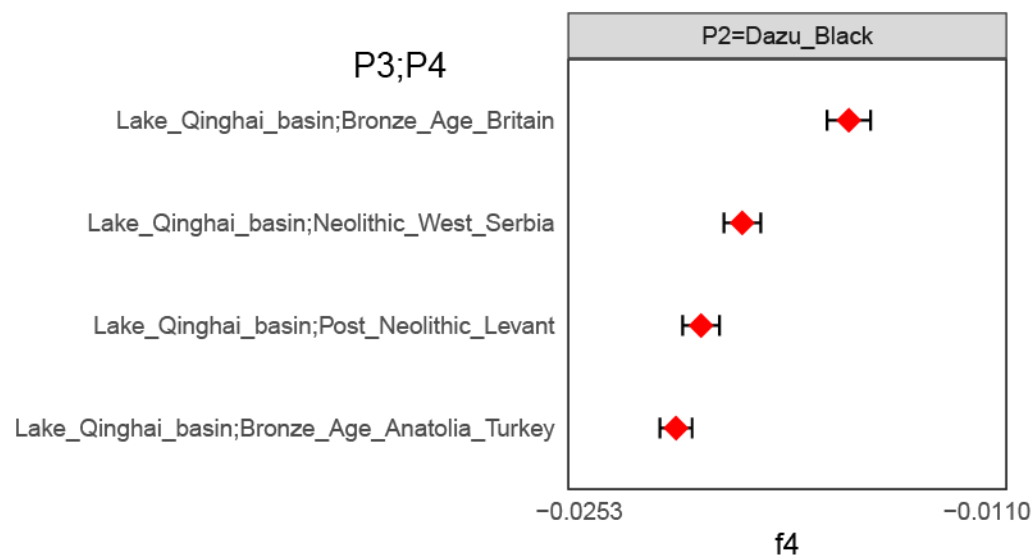
Supplementary Figure S3 The heatmap generated from the pairwise outgroup- f_3 analysis in this study.



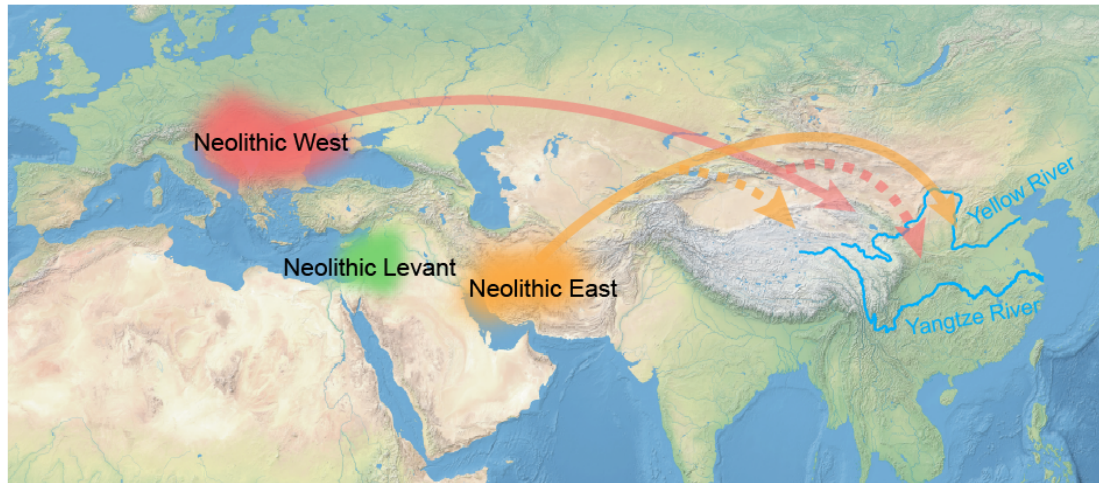
Supplementary Figure S4 f_4 -statistics suggest that the Lake_Qinghai_basin have closer genetic relationship with West ancestry than Neolithic Levant populations.



Supplementary Figure S5 f_4 -statistics indicate that the Lake_Qinghai_basin show close genetic relationship to Bronze_Age_Britain and Neolithic_West_Serbia than Bronze_Age_Anatolia_Turkey.



Supplementary Figure S6 f_4 -statistics show the closest genetic relationship between Dazhu black goats and ancient Lake_Qinghai_basin goats.



Supplementary Figure S7 The postulated dispersal routes of Chinese goat.

Supplementary Tables

Supplementary Table S1 Modern goat datasets used in this study

Supplementary Table S2 All sample information, related to Table 1

Supplementary Table S3 Relationship inference by lcmlkin. No pair of individuals closer than third-degree was detected

Supplementary Table S4 Result of f_4 test in the format of $f_4(\text{bezoar}, \text{Lake_Qinghai_basin}; \text{Neolithic East group}, \text{Neolithic West group})$. $|Z| > 3$ indicates the significant values, highlighted in Data by red (negative) and blue (positive); $2 < |Z| < 3$ filled with orange (negative) and light blue (positive)

Supplementary Table S5 Result of f_4 test in the format of $f_4(\text{bezoar}, \text{Lake_Qinghai_basin}; \text{Post_Neolithic_Levant}, \text{Neolithic West group})$

Supplementary Table S6 Result of f_4 test in the format of $f_4(\text{bezoar}, \text{Lake_Qinghai_basin}; \text{Bronze_Age_Anatolia_Turkey}, \text{Neolithic_West_Serbia/Bronze_Age_Britain})$

Supplementary Table S7 Result of f_4 test in the format of $f_4(\text{bezoar}, \text{Bronze_Age_Britain/Neolithic_West_Serbia}; \text{other Neolithic ancient goat population}, \text{Lake_Qinghai_basin})$

Supplementary Table S8 Read counts supporting each allele variant at the altitude adaptation trait

Supplementary Table S9 Result of f_4 test in the format of $f_4(\text{bezoar}, \text{present-day Chinese goats}; \text{Lake_Qinghai_basin}, \text{Neolithic East Group})$

Supplementary Table S10 Result of f_4 test in the format of $f_4(\text{bezoar}, \text{Dazu_Black}; \text{Lake_Qinghai_basin}, \text{Neolithic West group/Post_Neolithic_Levant})$

Supplementary Table S11 Result of f_4 test in the format of f_4 (bezoar, Neolithic East group, Dazu_Black; Lake_Qinghai_basin)

Supplementary Table S12 Result of f_4 test in the format of f_4 (bezoar, Lake_Qinghai_basin, present-day Chinese goats; Neolithic East group)

Supplementary Table S13 f_4 -ratio modelling result of modern Chinese goat populations

Supplementary Tables S1-S13 are listed as a separate Excel file due to their large size.